3. What rule of inference is used in each of these arguments?

a) Alice is a mathematics major. Therefore, Alice is either a mathematics major or a computer science major.

Soln: p: Alice is a mathematics major q: Alice is a computer science major.

The form of the argument is: $\dfrac{p}{p \vee q}$. Rule of inference: $\boxed{\text{Addition}}$

b) Jerry is a mathematics major and a computer science major. Therefore, Jerry is a mathematics major.

Soln: p: Jerry is a mathematics major q: Jerry is a computer science major. The form of the argument is: $\dfrac{p \wedge q}{p}$.

Rule of inference: $\boxed{\text{Simplification}}$

c) If it is rainy, then the pool will be closed. It is rainy. Therefore the pool is closed.

Soln: p: It is rainy q: The pool is closed.

The form of the argument is: $\dfrac{\begin{matrix} p \to q \\ p \end{matrix}}{q}$ Rule of inference: $\boxed{\text{Modus ponens}}$

d) If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.

Soln: p: It will snow today q: The university is closed today.

The form of the argument is: $\dfrac{\begin{matrix} p \to q \\ \neg q \end{matrix}}{\neg p}$ Rule of inference: $\boxed{\text{Modus tollens}}$

e) If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.